CSCI218: Foundations of Artificial Intelligence

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Outline

- What is AI?
- Risks and Benefits of AI
- Agents and Environments
- The Concept of Rationality
- The Nature of Environments
- The Structure of Agents

What is AI?

- Researchers have pursued several different versions of AI
- Two dimensions
  - Human vs. rational
  - Thought vs. behavior
- Lead to four approaches

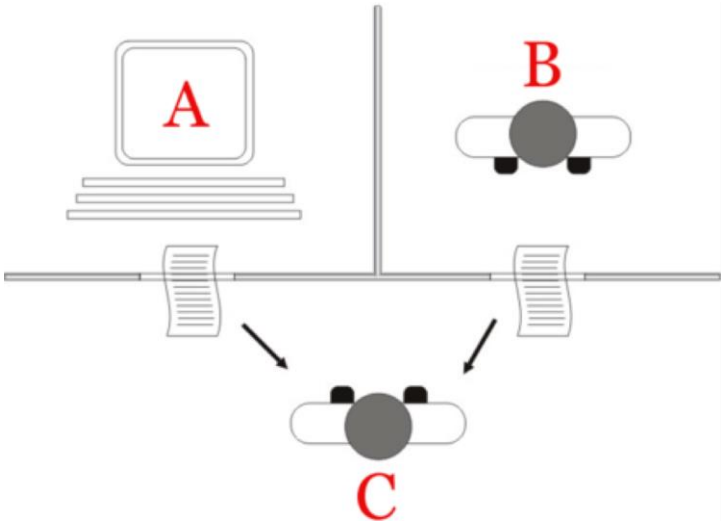
Acting humanly **The Turing test approach**	**Thinking humanly** **The cognitive modeling approach**
Acting rationally **The rational agent approach**	**Thinking rationally** **The "laws of thought" approach**

What is AI?

Acting humanly: The **Turing test** approach

- What capabilities are needed?
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning

  - Computer vision
  - Speech recognition
  - Robotics



What is AI?

Thinking humanly: The **cognitive modeling** approach

- We can learn about human thought in three ways:
  - introspection—trying to catch our own thoughts as they go by;
  - psychological experiments—observing a person in action;
  - brain imaging—observing the brain in action.
- Cognitive science
  - Computer models from AI + experimental techniques from psychology
  - Construct precise and testable theories of the human mind.
- Read minds (AI + Cognitive science)

What is AI?

Thinking rationally The "**laws of thought**" approach

- Syllogisms from Aristotle
  - "*Socrates is a man and all men are mortal and concludes that Socrates is mortal.*"
- Logic and certainty

Connective symbols	Word	Technical term	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only if	Biconditional	$A \Leftrightarrow B$
$\neg$ or $\sim$	Not	Negation	$\neg A$ or $\neg B$

- Probability and uncertainty

What is AI?

Acting rationally: The **rational agent** approach

- **Agent**: something that perceives and acts
  - Computer programs
  - Computer agents
  - Abstractly, an agent is a function from percept histories to actions:

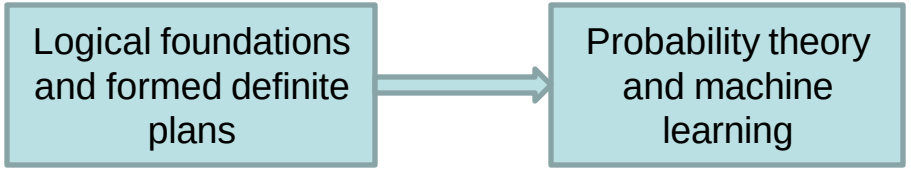
$$f : P^* \rightarrow A$$

- A **rational** agent
  - is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- Two advantages over the other approaches
  - More general + More amenable

What is AI?

The rational-agent approach to AI has prevailed

- AI has focused on
  - The study and construction of agents that do the right thing
  - The right thing is defined by the objective that we provide to the agent

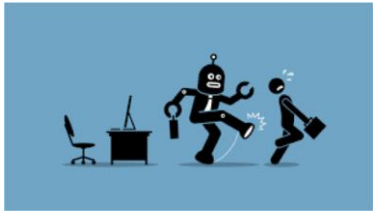


- The standard model
  - Perfect rationality vs. Limited rationality
  - Its inadequacy

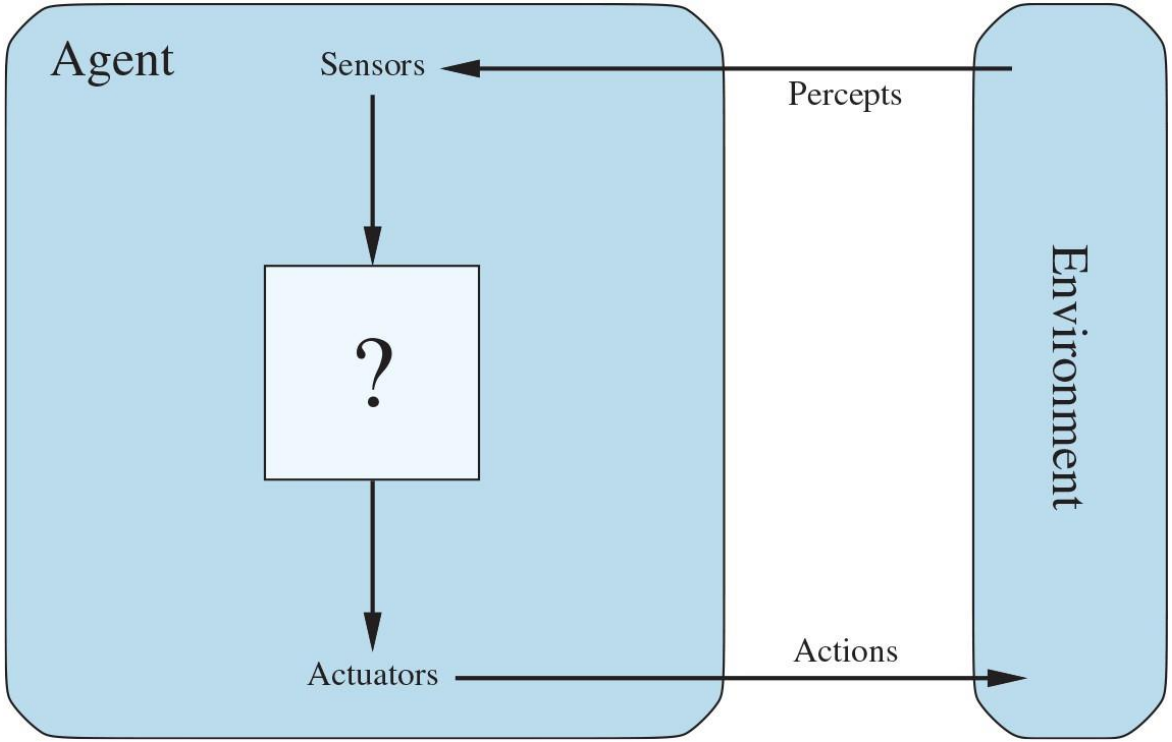
Risks and Benefits of AI

--"*Mechanical arts are of ambiguous use, serving as well for hurt as for remedy*"

- Benefits of using AI
  - Free humanity from work; Increase goods; Accelerate research, etc.
- Risks of misuse of AI
  - Weapons, surveillance, biasness, employment, safety, security, etc.
- As AI becomes more capable, they will take more roles
- The importance of governance and, eventually, regulation.



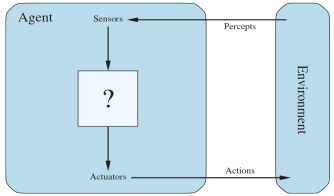
Agents and environments



Agents and environments

- An agent ***perceives*** its environment through ***sensors*** and ***acts*** upon it through ***actuators***
- The ***agent function*** maps percept sequences to actions

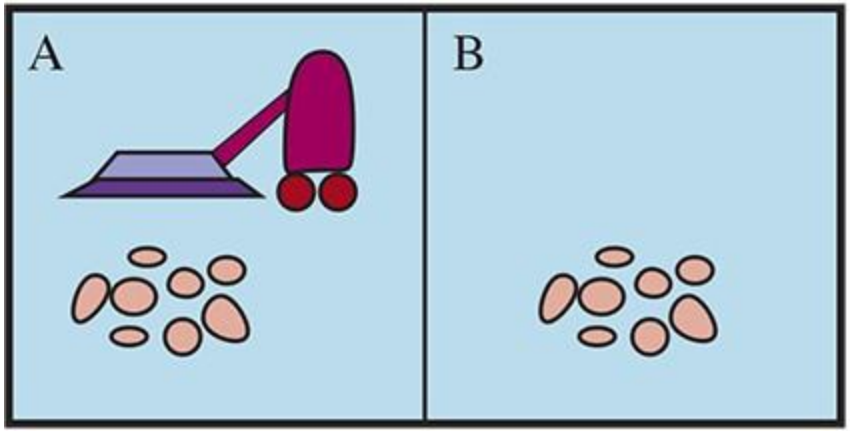
$$f : \mathrm{P}^* \rightarrow \mathrm{A}$$



- It is generated by an ***agent program*** running on a ***machine***
- What is the **environment**?
  - Part of the universe whose state we care about when designing this agent
  - It affects what the agent perceives
  - It is affected by the agent's actions.

Agents and environments

- A vacuum-cleaner world with just two locations.
  - Each location can be clean or dirty
  - The agent can move left or right and can clean the square that it occupies.



- What is the environment?
- What are the percepts?
- What are the actions?
- What is the agent function?
- What makes the vacuum-cleaner good or bad?

A Vacuum-Cleaner Agent

Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean]$, $[A, Clean]$	*Right*
$[A, Clean]$, $[A, Dirty]$	*Suck*
.	.

```
function Reflex-Vacuum-Agent([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

What is the right function?

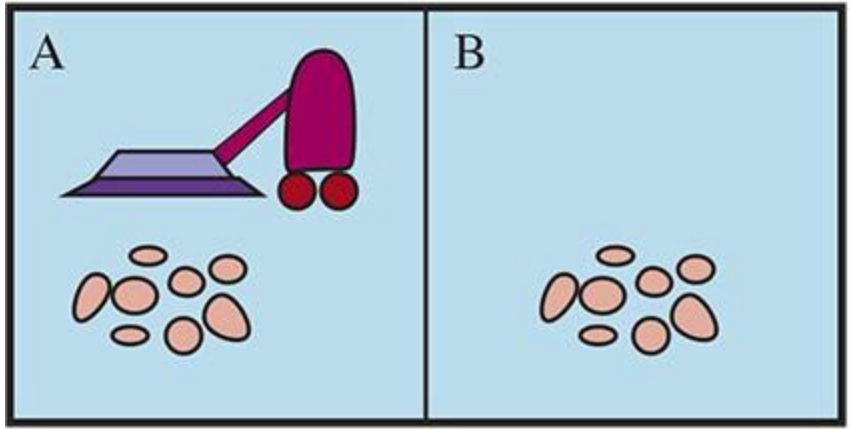
Can it be implemented in a small agent program?

Good Behavior: The Concept of Rationality

- A rational agent --- one that does the right thing.
  - What does it mean to do the **right** thing?
- Consequentialism
  - We evaluate an AI agent's behavior by its **consequences**.
- Perception ➔ Action ➔ Environment ➔ Change its state
  - Is the resulted sequence of environment states **desirable**?
- This notion of desirability
  - Captured by a **performance measure**
  - It evaluates any given sequence of environment states.
- Performance measure is in the mind of the designer / users of the machine
  - It can be quite hard to formulate a performance measure correctly

Good Behavior: The Concept of Rationality

*As a general rule, it is better to design performance measures according to **what one actually wants to be achieved in the environment**, rather than according to **how** one thinks the agent should behave.*



- Measure its performance by the amount of dirt cleaned up in a single eight-hour shift?
- Or, reward the agent for having a clean floor?

- In this subject, we usually assume that the **performance measure can be specified correctly** (although it is not always this case)

Good Behavior: The Concept of Rationality

- Wolves and sheep are placed at random
- The wolves need to catch all the sheep in 20 seconds while avoiding some boulders.
- A **point system** is implemented
  - A wolf caught a sheep: +10 points
  - Hit a boulder: -1 point
  - -0.1 point per second passed to encourage wolves to catch the sheep quickly.
- **Goal** is to train the AI wolves to figure out a way to maximize its scores.

Good Behavior: The Concept of Rationality

- **Rational** depends on four things:
  - The **performance measure** that defines the criterion of success.
  - The agent's prior knowledge of the **environment**.
  - The **actions** that the agent can perform.
  - The agent's **percept** sequence to date.

- A definition of **a rational agent**:
  - For each possible **percept** sequence, a rational agent should select an **action** that is expected to maximize its **performance measure**, given the evidence provided by the **percept** sequence and whatever built-in **knowledge** the agent has.

Good Behavior: The Concept of Rationality

- Difference between “rationality” and “**omniscience**”
  - An omniscient agent knows the **actual** outcome of its actions and can act accordingly
  - Rationality is not the same as **perfection**.
  - Rationality maximizes **expected** performance, while perfection maximizes **actual** performance
  - Rational choice depends only on the percept sequence **to date**.

- Our definition requires a rational agent not only to **gather information** but also to **learn** as much as possible from what it perceives.

- A rational agent should be **autonomous**—it should learn what it can to compensate for partial or incorrect prior knowledge

The Task Environments

- Task environments
  - The “problems” to which rational agents are the “solutions.”
- PEAS
  - **Performance, Environment, Actuators, Sensors**
- We must always be to specify the task environment as fully as possible.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Internet Shopping Agent

Performance measure??

- price, quality, appropriateness, efficiency

Environment??

- current and future WWW sites, vendors, shippers

Actuators??

- display to user, follow URL, fill in form

Sensors??

- HTML pages (text, graphics, scripts)

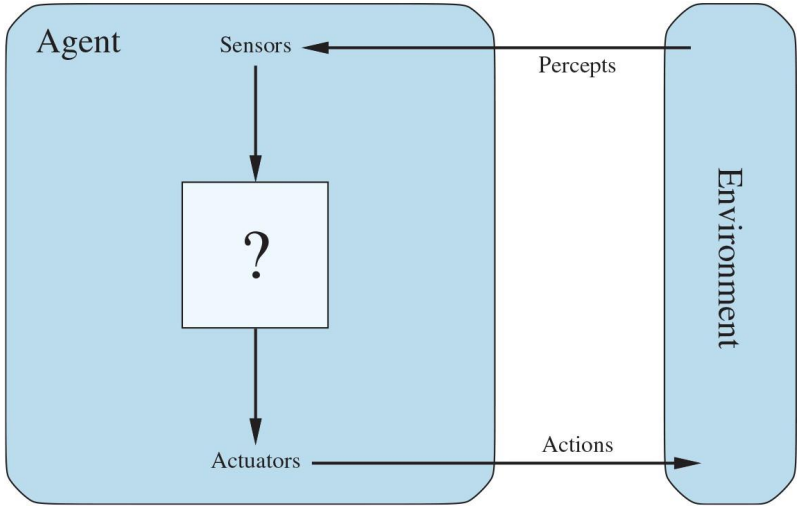
The Task Environments

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

The Nature of Environments

Properties of **task environments**

- Fully observable vs. partially observable
- Single-agent vs. multiagent
- Deterministic vs. nondeterministic.
- Episodic vs. sequential
- Static vs. dynamic
- Discrete vs. continuous
- Known vs. unknown



The Nature of Environments

The **hardest** case is *partially observable, multiagent, nondeterministic, sequential, dynamic, continuous, and unknown*.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

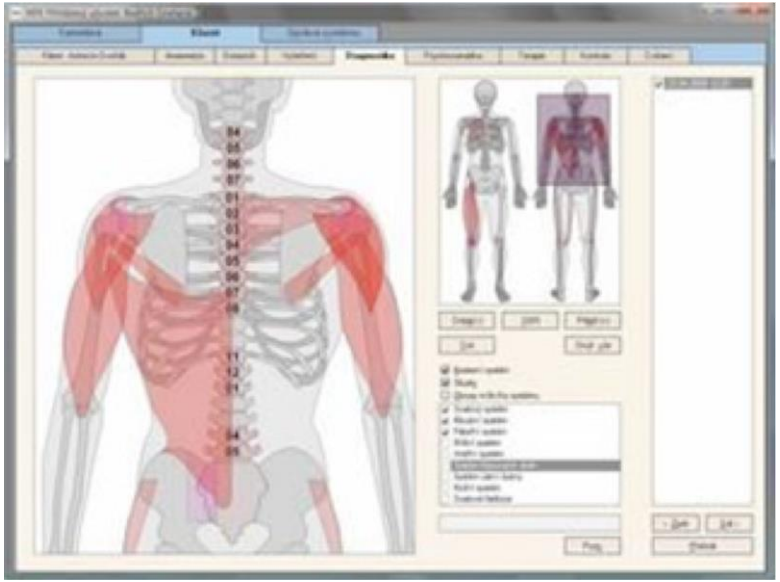
PEAS: Automated taxi

- Performance measure
  - Income, happy customer, vehicle costs, fines, insurance premiums
- Environment
  - US streets, other drivers, customers, weather, police…
- Actuators
  - Steering, brake, gas, display/speaker
- Sensors
  - Camera, radar, accelerometer, engine sensors, microphone, GPS

Image: http://nypost.com/2014/06/21/how-google-might-put-taxi-drivers-out-of-business/

PEAS: Medical diagnosis system

- Performance measure
  - Patient health, cost, reputation
- Environment
  - Patients, medical staff, insurers, courts
- Actuators
  - Screen display, email
- Sensors
  - Keyboard/mouse



The Structure of Agents

- What is the **inside** of an agent?
- Recall the job of AI
  - design an agent program that implements the **agent function**—the mapping from **percepts** to **actions**.

agent = agent architecture + agent program

- Agent architecture
  - some computing device with physical sensors and actuators

The Structure of Agents

- We mainly focus on “agent program” in this subject
- Agent program
  - Input: the current percept from the sensors
  - Output: an action to the actuators.
- What if the agent’s actions need to depend on the entire percept sequence?
  - Then the agent will have to remember the percepts.

The Structure of Agents

- Let's look at a rather trivial agent program
  - It keeps track of the percept sequence,
  - and then uses it to index into a table of actions to decide what to do.

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

- However, a lookup table based approach does not scale

The Structure of Agents

- The key challenge for AI is to
  - Produce rational behavior from a smallish program rather than from a vast table.
- Four basic kinds of agent programs
  - Simple reflex agents;
  - Model-based reflex agents;
  - Goal-based agents; and
  - Utility-based agents.

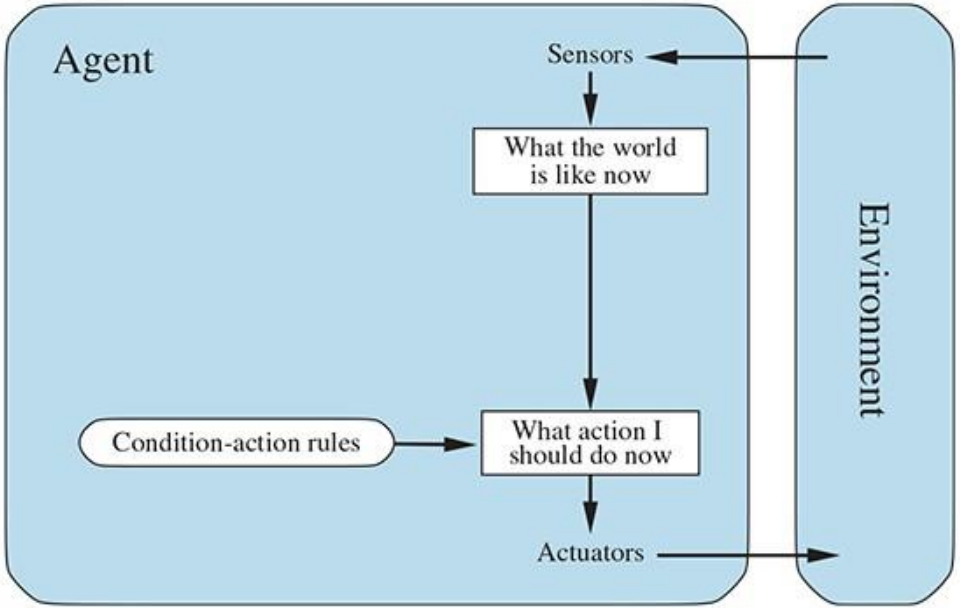
The Structure of Agents

Simple reflex agents

- They select actions on the basis of the **current percept**, ignoring the rest of the percept history.
- Leaned reflex and innate reflex

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

The Structure of Agents




```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

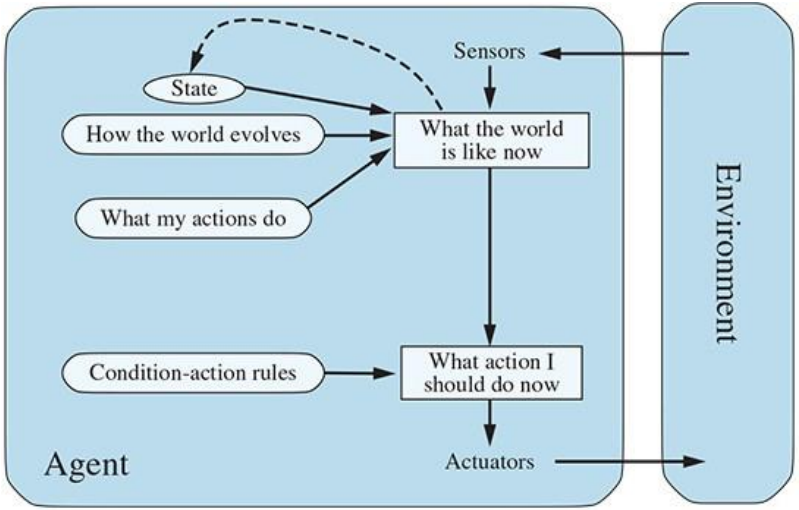
- Simple reflex agents are **simple** but have **limited** intelligence.
- They will work only if the correct decision can be made on the basis of just the current percept—that is, **only if** the environment is **fully observable**.

The Structure of Agents

Model-based reflex agents

- To handle partial observability, the agent needs to keep track of the part of the world it can't see now.
  - The agent should maintain some sort of **internal state** that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
- Updating this internal state requires
  - **Transition model**: how the world changes over time,
  - **Sensor model**: how the state of the world is reflected in the agent's percepts.

The Structure of Agents




```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              transition_model, a description of how the next state depends on
                    the current state and action
              sensor_model, a description of how the current world state is reflected
                    in the agent's percepts
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

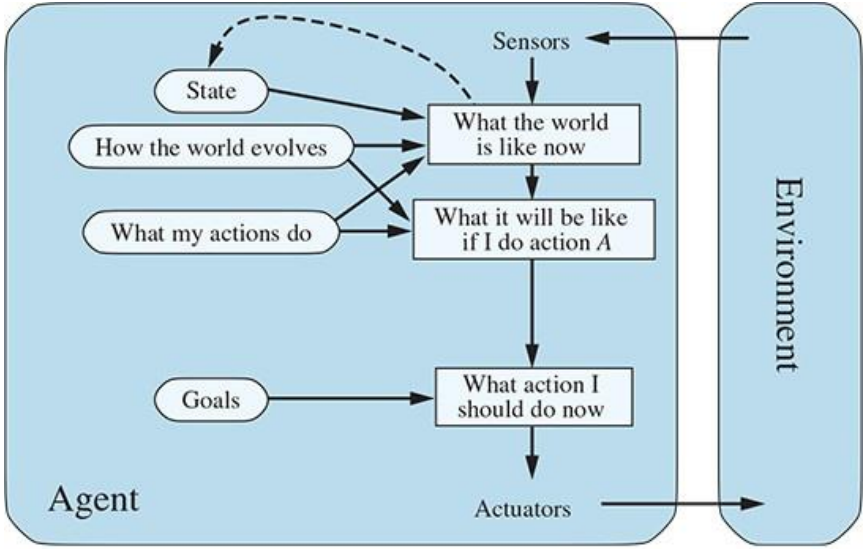
- A model-based reflex agent keeps track of the current state of the world, using an **internal model**. It then chooses an action in the same way as the reflex agent.

The Structure of Agents

Goal-based agents

- The agent needs some sort of **goal** information that describes situations that are desirable
  - It combines the **goal** information with the model to choose **actions** that achieve the goal.
- Goal-based action selection could be straightforward or tricky
  - Search and planning to find action sequences to achieve a goal.
- More flexible because the knowledge that supports its decisions is represented explicitly and can be modified

The Structure of Agents



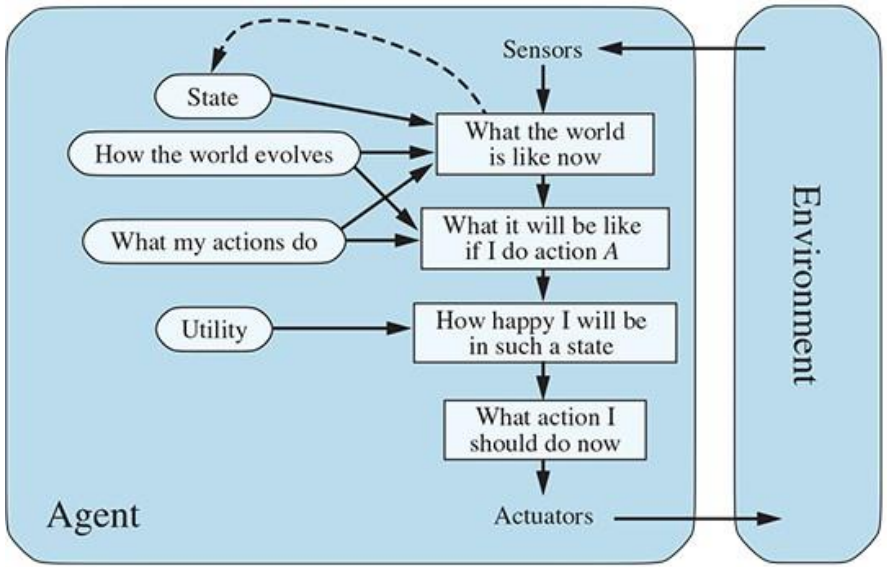
A model-based, goal-based agent keeps track of the world state as well as a set of **goals** it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

The Structure of Agents

Utility-based agents

- Allow a comparison of different actions in the eye of the agent.
- An agent's **utility function** is essentially an internalization of the **performance measure**.
  - Provided that the internal utility function and the external performance measure are in agreement, an agent that chooses actions to **maximize its utility** will be rational according to the external performance measure.
- Has many advantages in terms of flexibility
  - Conflicting goals, uncertainty, **expected utility**

The Structure of Agents



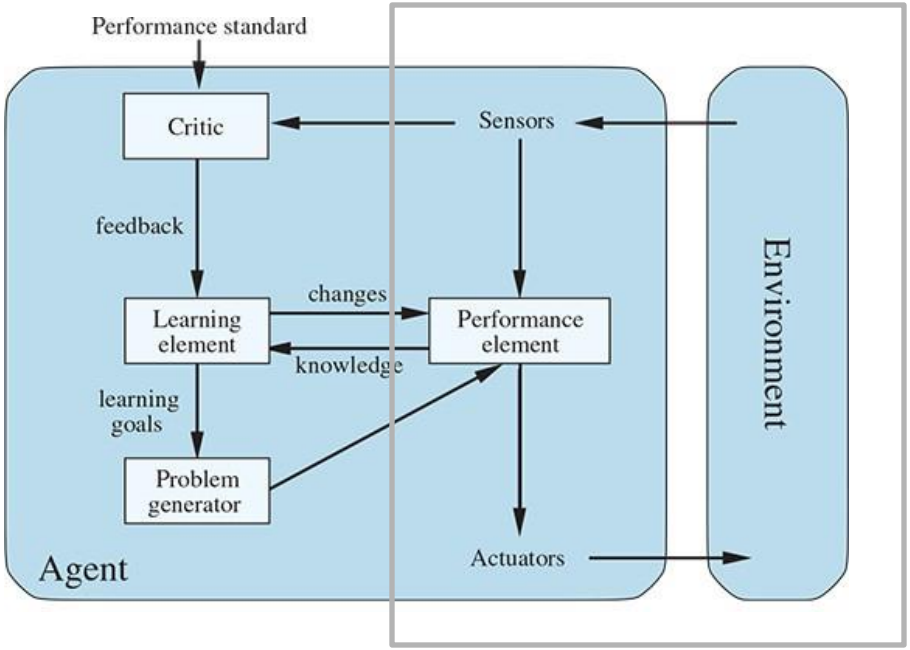
- A model-based, utility-based agent uses a model of the world, along with a **utility function** that measures its preferences among states of the world.
- It chooses the action that leads to the best **expected utility**, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

The Structure of Agents

Agents that are able to **learn**

- How the agent programs "come into being?"
- Build a machine that can **learn** and then **teach** them
  - The preferred method for creating state-of-the-art systems.
  - It allows the agent to operate in initially unknown environments and to become more competent than its initial knowledge alone might allow.
- A learning agent involves four conceptual components
  - learning element, critic
  - performance element, problem generator

The Structure of Agents



Learning in intelligent agents can be summarized as a process of **modification of each component of the agent** to bring the components into closer agreement with the available feedback information, thereby **improving the overall performance** of the agent.

- A general **learning** agent. The "performance element" box represents what we have previously considered to be the whole agent program. Now, the "learning element" box gets to modify that program to improve its performance.

Summary of Agents

- Agents interact with environments through actuators and sensors
- The agent function describes what the agent does in all circumstances
- The performance measure evaluates the environment sequence
- A perfectly rational agent maximizes expected performance
- Agent programs implement (some) agent functions
- PEAS descriptions define task environments
- Environments are categorized along several dimensions:

  observable? deterministic? episodic? static? discrete? single-agent?

Several basic agent architectures exist:

reflex, reflex with state (model-based reflex), goal-based, utility-based, learning-based

Summary

- What is AI?
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Questions?

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